

EDUPORTAL

AWS Powered Academic Management Portal

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1. Abstract

ScholarPoint is a cloud-based Academic Management Portal developed for the Electronics & Computer Engineering Department at Parikrama College of Engineering. The project demonstrates real-world deployment of a PHP web application on Amazon Web Services (AWS) infrastructure.

The portal provides separate interfaces for Teachers and Students. Teachers can manage student approvals, assign classes (SE/TE/BE), manage subjects, post announcements, schedule meetings, and monitor attendance. Students can access study materials, assignments, and class schedules after teacher approval.

The application is deployed on AWS using EC2, RDS, VPC, ALB, and CloudWatch — demonstrating production-grade cloud architecture suitable for educational institutions.

2. Project Overview

2.1 Project Details

Project Name	Eduportal— AWS Powered Academic Management Portal
Technology Stack	PHP, MySQL, HTML, CSS, JavaScript
Cloud Platform	Amazon Web Services (AWS)
Web Server	Apache2 on Ubuntu 22.04
Database	AWS RDS MySQL 8.0
Region	ap-south-1 (Mumbai)
Department	Electronics & Computer Engineering
Target Users	SE, TE, BE Students and Teachers

2.2 Key Features

- Student registration with class selection (SE/TE/BE) and teacher approval system
- Teacher dashboard with pending approvals, class assignment, and user management
- Subject management with teacher-to-subject assignment
- Study materials upload and download
- Homework/Assignment submission
- Online meetings via Jitsi Meet integration
- Announcements and notifications

3. System Architecture Diagram

Internet



Application Load Balancer (ALB)



EC2 (Apache + PHP) — Public Subnet



RDS MySQL 8.0 — Private Subnet

3-Tier Architecture deployed on AWS:

- Presentation Layer: ALB + EC2 (Apache/PHP)
- Application Layer: PHP Business Logic
- Database Layer: AWS RDS MySQL in Private Subnet

4. AWS Services Used

AWS Service	Purpose	Benefit
Amazon EC2	Hosts PHP web application on Ubuntu 22.04	Full control, scalable compute
Amazon RDS	Managed MySQL 8.0 database in private subnet	Automated backups, high availability
Amazon VPC	Isolated network with public/private subnets	Security, network segmentation
Application Load Balancer	Distributes traffic across EC2 instances	High availability, health checks
Amazon CloudWatch	Monitors EC2, RDS, ALB metrics and logs	Real-time alerts, log analysis
AWS IAM	Role-based access for EC2 to CloudWatch	Secure access management
Internet Gateway	Connects VPC to internet	Public subnet internet access
Security Groups	Firewall rules for EC2, RDS, ALB	Port-level traffic control

5. VPC Network Design

5.1 VPC Configuration

VPC Name	test-vpc
CIDR Block	12.0.0.0/16
Region	ap-south-1 (Mumbai)
Internet Gateway	Attached to VPC

5.2 Subnet Design

Subnet Name	CIDR	AZ	Type
public-subnet	12.0.1.0/24	ap-south-1a	Public (EC2)
private-subnet	12.0.2.0/24	ap-south-1a	Private (RDS)
public-subnet-2	12.0.4.0/24	ap-south-1b	Public (ALB)
private-subnet-2	12.0.3.0/24	ap-south-1b	Private (RDS)

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Asia Pacific (Mumbai)

Account ID: 4713-4100-2257

Sandip Raut

VPC > Your VPCs > vpc-0bed51868424d8a31

vpc-0bed51868424d8a31 / test-vpc

Actions

Details

VPC ID

vpc-0bed51868424d8a31

DNS resolution

Enabled

Main network ACL

acl-0896e03c350b17fc7

IPv6 CIDR (Network border group)

-

Encryption control ID

-

State

Available

Tenancy

default

Default VPC

No

Network Address Usage metrics

Disabled

Encryption control mode

-

Block Public Access

Off

DHCP option set

dopt-0c48d51c74ca74930

IPv4 CIDR

12.0.0.0/16

Route 53 Resolver DNS Firewall rule groups

-

DNS hostnames

Disabled

Main route table

rtb-0a4ed45416e37add0

IPv6 pool

-

Owner ID

471341002267

Resource map

CIDRs

Flow logs

Tags

Integrations

Resource map

Show all details

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Asia Pacific (Mumbai)

Account ID: 4713-4100-2257

Sandip Raut

VPC > Internet gateways > igw-0ace662220d204adc

Internet gateway igw-0ace662220d204adc successfully attached to vpc-0bed51868424d8a31

igw-0ace662220d204adc / IGW-test

Actions

Details

Internet gateway ID

igw-0ace662220d204adc

State

Attached

VPC ID

vpc-0bed51868424d8a31 | test-vpc

Owner

471341002267

Tags (1)

Search tags

Key

Value

Name

IGW-test

Manage tags

< 1 >

6. Database Design (RDS)

6.1 RDS Configuration

DB Identifier	scholarpoint-db
Engine	MySQL 8.0
Instance Class	db.t3.micro (Free Tier)
Storage	20 GB gp2
Subnet Group	scholarpoint-rds-subnet-group
Public Access	No (Private Subnet)
Multi-AZ	No (Single-AZ)
Automated Backups	1 day retention

6.2 Database Tables

- users — Student and Teacher accounts with role, class, and approval status
- classes — SE, TE, BE class definitions
- subjects — Subject list with class mapping
- teacher_subjects — Teacher to subject assignment mapping
- announcements — Teacher announcements
- assignments — Student homework submissions
- attendance — Student attendance records
- meetings — Scheduled online meetings
- materials — Study material files
- exam_marks — Student examination marks

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Asia Pacific (Mumbai)

Account ID: 4715-4100-2287

Sandip R

> scholarpoint-db

scholarpoint-db



Modify

Actions

Summary

DB identifier scholarpoint-db	Status Available	Role Instance	Engine MySQL Community	Recommendations
CPU -	Class db.t3.micro	Current activity	Region & AZ ap-south-1a	

Connectivity & security

Monitoring

Logs & events

Configuration

Zero-ETL integrations

Maintenance & backups

Data migrations

Connect using

Info

☒ Code snippets
Use when connecting through SDK, APIs, or third-party tools including agents.

☐ CloudShell
Use for a quick access to AWS CLI that launches directly from the AWS Management Console.

☐ Endpoints
Use when connecting through any IDE interface.

Internet access gateway
Disabled

IAM Authentication
Disabled

Programming language
MySQL (macOS)

Endpoint type
Instance endpoint



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Aurora and RDS

Subnet groups

Create DB subnet group

Create DB subnet group

To create a new subnet group, give it a name and a description, and choose an existing VPC. You will then be able to add subnets related to that VPC.

Subnet group details

Name
You won't be able to modify the name after your subnet group has been created.
scholarpoint-rds-subnet-group
Must contain from 1 to 255 characters. Alphanumeric characters, spaces, hyphens, underscores, and periods are allowed.

Description
Private subnets for ScholarPoint RDS

VPC
Choose a VPC identifier that corresponds to the subnets you want to use for your DB subnet group. You won't be able to choose a different VPC identifier after your subnet group has been created.
test-vpc (vpc-0bed51868424d8a31)
3 Subnets, 2 Availability Zones

Add subnets

Availability Zones
Choose the Availability Zones that include the subnets you want to add.

7. Load Balancer Configuration (ALB)

ALB Name	scholarpoint-alb
Type	Application Load Balancer
Scheme	Internet-facing
Availability Zones	ap-south-1a, ap-south-1b
Listener	HTTP:80 → scholarpoint-tg
Target Group	scholarpoint-tg (HTTP:80)
Health Check Path	/index.php
Health Check Interval	30 seconds
Target Status	Healthy

7.1 Security Groups

- scholarpoint-alb-sg: Allows HTTP:80 from Anywhere (0.0.0.0/0)
- EC2 Security Group: Allows HTTP:80 from scholarpoint-alb-sg only
- scholarpoint-rds-sg: Allows MySQL:3306 from EC2 Security Group only

Successfully created load balancer: scholarpoint-alb
It might take a few minutes for your load balancer to fully set up and route traffic. Targets will also take a few minutes to complete the registration process and pass initial health checks.

scholarpoint-alb

- Details**

Load balancer type Application	Status Provisioning	VPC vpc-Cbed5186d424d8a31	Load balancer IP address type IPv4
Scheme Internet-facing	Hosted zone ZP97RAFLXTNZK	Availability Zones subnet-0799da6f60023acbe ap-south-1b (aps1-a-z3) subnet-08ba990263fd85274 ap-south-1a (aps1-a-z1)	Date created May 25, 2026, 11:36 (UTC+05:30)
Load balancer ARN arn:aws:elasticloadbalancing:ap-south-1:471341002267:loadbalancer/app:scholarpoint-lb/4f5e0a03ef5e82c3		DNS name info scholarpoint-alb-352794007.ap-south-1.elb.amazonaws.com (A Record)	

Listeners and rules | Network mapping | Resource map | Security | Monitoring | Integrations | Attributes | Capacity | Tags

8. Monitoring — CloudWatch

8.1 Dashboard — ScholarPoint-Dashboard

- EC2 CPUUtilization — Real-time CPU usage graph
- ALB RequestCount — Incoming traffic monitoring
- RDS DatabaseConnections — Active DB connections
- EC2 NetworkIn/NetworkOut — Network traffic
- Memory Used % — Custom metric via CloudWatch Agent
- Disk Used % — Storage utilization monitoring

8.2 CloudWatch Agent

- Installed on EC2: amazon-cloudwatch-agent v1.300067
- Apache Error Logs → Log Group: ScholarPoint-Apache-Errors
- Apache Access Logs → Log Group: ScholarPoint-Apache-Access
- Custom Metrics: mem_used_percent, disk_used_percent
- IAM Role: EC2-CloudWatch-Role with CloudWatchAgentServerPolicy

9. Challenges & Solutions

Challenge	Solution
Login not working after deployment	Fixed md5 vs password_hash mismatch in register.php and login.php
HTTP 500 error on register.php	\$conn null due to wrong DB credentials in connect.php
RDS connection failed — Access Denied	Password mismatch — reset RDS master password via Modify
ALB showing Unhealthy target	Fixed security group — added ALB sg as source for EC2 HTTP:80 rule
CloudWatch Agent stopped	Fixed JSON config — metrics section was nested inside logs section

RDS subnet group required 2 AZs	Created private-subnet-2 in ap-south-1b with CIDR 12.0.3.0/24
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10. Future Scope

- Amazon S3 — Store study materials and assignments in S3 instead of EC2 local storage
- Auto Scaling Group — Automatically add EC2 instances during peak exam periods
- CloudWatch Alarms — Email alerts for CPU > 80%, RDS storage > 80%, server down
- HTTPS / SSL — Add domain name and SSL certificate via AWS Certificate Manager
- Amazon SES — Email notifications for student approvals, announcements
- AWS WAF — Web Application Firewall for additional security
- CI/CD Pipeline — Automated deployment using AWS CodePipeline and CodeDeploy